A M-scheme double commutators

The one-body piece of the double commutator is

$$
\begin{aligned}
f^I_{ij} =& \frac{1}{2}\sum_{abcde} \left(\bar{n}_a\bar{n}_b n_c n_d - n_a n_b \bar{n}_c\bar{n}_d\right) \\
& \times \left(\Omega_{cdab}\Omega_{abce}\Gamma_{eidj} + \Omega_{abcd}\Omega_{ceab}\Gamma_{diej}\right) \quad (1a)\\
f^{II}_{ij} =& \frac{1}{2}\sum_{abcde} \left(n_a n_b \bar{n}_c\bar{n}_e - \bar{n}_a\bar{n}_b n_c n_e\right) \\
& \times \left(\Gamma_{cdab}\Omega_{abce}\Omega_{eidj} + \Gamma_{abcd}\Omega_{ceab}\Omega_{diej}\right) \quad (1b)\\
f^{III_a}_{ij} =& \sum_{abcde} \left(\bar{n}_a\bar{n}_b n_c n_d - n_a n_b \bar{n}_c\bar{n}_d\right) \\
& \times \left(\Omega_{abcd}\Omega_{idae}\Gamma_{cejb} - \Omega_{abcd}\Omega_{edaj}\Gamma_{cieb}\right) \quad (1c)\\
f^{III_b}_{ij} =& \frac{1}{4}\sum_{abcde} \left(\bar{n}_a\bar{n}_b n_c n_d - n_a n_b \bar{n}_c\bar{n}_d\right) \\
& \times \left(\Omega_{abcd}\Omega_{cdej}\Gamma_{eiab} - \Omega_{abcd}\Omega_{eiab}\Gamma_{cdej}\right) \quad (1d)
\end{aligned}
$$

These may be re-expressed in a factorized form

$$
\begin{aligned}
f^I_{ij} &= \sum_{ab} \left(\chi^\alpha_{ab}\Gamma_{biaj} + \chi^\alpha_{ab}\Gamma_{aibj}\right) \quad (2a)\\
f^{II}_{ij} &= \sum_{ab} \left(\chi^\beta_{ab}\Omega_{biaj} - \chi^\beta_{ab}\Omega_{aibj}\right) \quad (2b)\\
f^{III_a}_{ij} &= \sum_{abc} \left(\chi^\gamma_{icab}\Gamma_{abjc} - \Gamma_{ciab}\chi^\gamma_{abcj}\right) \quad (2c)\\
f^{III_b}_{ij} &= \sum_{abc} \left(\chi^\delta_{ciab}\Gamma_{abcj} - \Gamma_{ciab}\chi^\delta_{abcj}\right) \quad (2d)
\end{aligned}
$$

where we have defined the intermediate quantities

$$
\begin{aligned}
\chi^\alpha_{ij} =& \frac{1}{2}\sum_{abc} \left(\bar{n}_a\bar{n}_b n_c n_i - n_a n_b \bar{n}_c\bar{n}_i\right)\Omega_{ciab}\Omega_{abcj} \quad (3a)\\
\chi^\beta_{ij} =& \frac{1}{2}\sum_{abc} \left(\bar{n}_a\bar{n}_b n_c n_i - n_a n_b \bar{n}_c\bar{n}_i\right)\Omega_{ciab}\Gamma_{abcj} \quad (3b)\\
\chi^\gamma_{ijkl} =& \sum_{ab} \left(n_a\bar{n}_b n_j\bar{n}_k - \bar{n}_a n_b\bar{n}_j n_k\right)\Omega_{ajkb}\Omega_{ibal} \quad (3c)\\
\chi^\delta_{ijkl} =& \frac{1}{4}\sum_{ab} \left(\bar{n}_i\bar{n}_j n_a n_b - n_i n_j\bar{n}_a\bar{n}_b\right)\Omega_{ijab}\Omega_{abkl} \quad (3d)
\end{aligned}
$$

We use superscript Greek letters to distinguish the intermediates.

The two-body piece of double the commutator is

$$
\begin{aligned}
\Gamma^I_{ijkl} =& \frac{1}{2}\sum_{abcd} \left(\bar{n}_a\bar{n}_b n_c + n_a n_b \bar{n}_c\right) \\
& \times \Big\{ (1-\hat{P}_{ij})\Omega_{ciab}\Omega_{abcd}\Gamma_{djkl}
\end{aligned}
$$

$$+ (1-\hat{P}_{kl})\Omega_{cdab}\Omega_{abcl}\Gamma_{ijkd}\Big\} \tag{4a}$$

$$\begin{aligned}\Gamma^{II}_{ijkl} =& -\frac{1}{2}\sum_{abcd}(\bar{n}_a\bar{n}_b n_c + \bar{n}_c n_a n_b) \\ &\times\Big\{(1-\hat{P}_{ij})\Omega_{cjab}\Gamma_{abcd}\Omega_{idkl} \\ &+ (1-\hat{P}_{kl})\Gamma_{cdab}\Omega_{abcl}\Omega_{ijkd}\Big\}\end{aligned} \tag{4b}$$

$$\begin{aligned}\Gamma^{III_a}_{ijkl} =& -\sum_{abcd}(\bar{n}_c\bar{n}_d n_a + n_a\bar{n}_c\bar{n}_d) \\ &\times\Big\{(1-\hat{P}_{ij})\Omega_{ajcd}\Omega_{idab}\Gamma_{cbkl} \\ &+ (1-\hat{P}_{kl})\Omega_{cdka}\Omega_{bacl}\Gamma_{ijbd}\Big\}\end{aligned} \tag{4c}$$

$$\begin{aligned}\Gamma^{III_b}_{ijkl} =& -\sum_{abcd}(\bar{n}_b n_c n_d + n_b\bar{n}_c\bar{n}_d)(1-\hat{P}_{ij})(1-\hat{P}_{kl}) \\ &\times(\Omega_{dcbk}\Omega_{biac}\Gamma_{jald} + \Omega_{jcbd}\Omega_{balc}\Gamma_{diak})\end{aligned} \tag{4d}$$

$$\begin{aligned}\Gamma^{III_c}_{ijkl} =& -\frac{1}{2}\sum_{abcd}(\bar{n}_a\bar{n}_b n_c + n_a n_b\bar{n}_c)(1-\hat{P}_{ij})(1-\hat{P}_{kl}) \\ &\times(\Omega_{abcl}\Omega_{idab}\Gamma_{cjkd} + \Omega_{icab}\Omega_{abdl}\Gamma_{djkc})\end{aligned} \tag{4e}$$

$$\begin{aligned}\Gamma^{IV_a}_{ijkl} =& -\sum_{abcd}(\bar{n}_c\bar{n}_d n_a + n_c n_d\bar{n}_a) \\ &\times\Big\{(1-\hat{P}_{ij})\Omega_{aicd}\Omega_{dbkl}\Gamma_{jcba} \\ &+ (1-\hat{P}_{kl})\Omega_{dcak}\Omega_{ijcb}\Gamma_{bald}\Big\}\end{aligned} \tag{4f}$$

$$\begin{aligned}\Gamma^{IV_b}_{ijkl} =& (1-\hat{P}_{ij})(1-\hat{P}_{kl})\sum_{abcd}(\bar{n}_a n_b\bar{n}_c + n_a\bar{n}_b n_c) \\ &\times(\Omega_{bica}\Omega_{jcld}\Gamma_{dabk} + \Omega_{cabk}\Omega_{jdlc}\Gamma_{bida})\end{aligned} \tag{4g}$$

$$\begin{aligned}\Gamma^{IV_c}_{ijkl} =& \frac{1}{2}(1-\hat{P}_{ij})(1-\hat{P}_{kl})\sum_{abcd}(\bar{n}_a\bar{n}_b n_d + n_a n_b\bar{n}_d) \\ &\times(\Omega_{abld}\Omega_{djck}\Gamma_{icab} + \Omega_{idab}\Omega_{cjdk}\Gamma_{ablc})\end{aligned} \tag{4h}$$

These may be expressed in a factorized form

$$\Gamma^{I}_{ijkl} = \sum_{a}\Big\{(1-\hat{P}_{ij})\chi^{\epsilon}_{ia}\Gamma_{ajkl} + (1-\hat{P}_{kl})\chi^{\epsilon}_{ak}\Gamma_{ijal}\Big\} \tag{5a}$$

$$\Gamma^{II}_{ijkl} = \sum_{a}\Big\{(1-\hat{P}_{ij})\chi^{\zeta}_{aj}\Omega_{iakl} - (1-\hat{P}_{kl})\chi^{\zeta}_{ak}\Omega_{ijal}\Big\} \tag{5b}$$

$$\Gamma^{III_a}_{ijkl} = -\sum_{ab}\{(1-\ hatP_{ij})\chi^{\eta}_{ijab}\Gamma_{abkl} + (1-\hat{P}_{kl})\Gamma_{ijab}\chi^{\eta}_{abkl}\} \tag{5c}$$

$$\Gamma^{III_b}_{ijkl} = -(1-\hat{P}_{ij})(1-\hat{P}_{kl})\sum_{ab}\Big(\chi^{\eta}_{bkai}\Gamma_{jbla} + \chi^{\eta}_{lajb}\Gamma_{aibk}\Big) \tag{5d}$$

$$\Gamma^{III_c}_{ijkl} = -\frac{1}{2}\Big(1-\hat{P}_{ij}\Big)(1-\hat{P}_{kl})\sum_{ab}\chi^{\theta}_{iabl}\Gamma_{bjka} \tag{5e}$$

$$\Gamma^{IV_a}_{ijkl} = -\sum_{ab}\Big\{(1-\hat{P}_{ij})\chi^{\kappa}_{ijab}\Omega_{bakl} - (1-\hat{P}_{kl})\Omega_{ijab}\chi^{\kappa}_{klba}\Big\} \tag{5f}$$

$$\Gamma^{IV_b}_{ijkl} =(1-\hat{P}_{ij})(1-\hat{P}_{kl})\sum_{ab}(\chi^{\iota}_{aibk}\Omega_{jbla} - \chi^{\iota}_{akbi}\Omega_{jalb}) \tag{5g}$$

$$\Gamma^{IV_c}_{ijkl} =\frac{1}{2}(1-\hat{P}_{ij})(1-\hat{P}_{kl})\sum_{ab}\chi^{\lambda}_{ialb}\Omega_{bjak} \tag{5h}$$

where we have defined additional intermediates

$$\chi^{\epsilon}_{ij} =\frac{1}{2}\sum_{abc}(\bar{n}_a\bar{n}_b n_c + n_a n_b \bar{n}_c)\,\Omega_{ciab}\Omega_{abcj} \tag{6a}$$

$$\chi^{\zeta}_{ij} =\frac{1}{2}\sum_{abc}(n_a n_b \bar{n}_c + \bar{n}_a\bar{n}_b n_c)\,\Gamma_{aibc}\Omega_{bcaj} \tag{6b}$$

$$\chi^{\eta}_{ijkl} =\sum_{ab}(\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k)\,\Omega_{iabl}\Omega_{bjka} \tag{6c}$$

$$\begin{aligned}\chi^{\theta}_{ijkl} =\sum_{ab}(n_a n_b \bar{n}_k &+ \bar{n}_a\bar{n}_b n_k \\ &+ n_a n_b \bar{n}_j + \bar{n}_a\bar{n}_b n_j)\,\Omega_{ijab}\Omega_{abkl}\end{aligned} \tag{6d}$$

$$\chi^{\iota}_{ijkl} =\sum_{ab}(\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k)\,\Omega_{bika}\Gamma_{iabl} \tag{6e}$$

$$\chi^{\kappa}_{ijkl} =\sum_{ab}(\bar{n}_a n_b n_k + n_a \bar{n}_b \bar{n}_k)\,\Omega_{ajbl}\Gamma_{ibka} \tag{6f}$$

$$\begin{aligned}\chi^{\lambda}_{ijkl} =\sum_{ab}\big\{&(\bar{n}_a\bar{n}_b n_l + n_a n_b \bar{n}_l)\,\Gamma_{ijab}\Omega_{abkl} \\ &+ (\bar{n}_a\bar{n}_b n_j + n_a n_b \bar{n}_j)\,\Omega_{ijab}\Gamma_{abkl}\big\}\end{aligned} \tag{6g}$$

We note that the intermediates χ do not in general have Hermitian symmetry or permutation symmetry under exchange of indices.

B J-coupled factorization version of double commutators

To take advantage of rotational symmetry, we work with J-coupled matrix elements. We work with un-normalized coupled states so that

$$A^{JM}_{ijkl} = \sum_{m_i m_j m_i m_l}\mathcal{C}^{JM}_{j_i m_i, j_j m_j}\mathcal{C}^{JM}_{j_k m_k, j_l m_l} A_{ijkl} \tag{7}$$

where the $\mathcal{C}$ are Clebsch-Gordan coefficients, and the uncoupled matrix element A_{ijkl} depends on the projections m_i, m_j, m_k, m_l. For the rotationally invariant operators considered in this work, A^{JM}_{ijkl} is independent of the total projection M and so M is not explicitly indicated. These expressions were obtained with the aid of the `amc` code [?]. In several cases, the expressions are simplified and made amenable to matrix multiplication if we use different coupling schemes. In addition to the standard coupling scheme, we employ both Pandya-transformed and cross-coupled matrix elements, illustrated in Fig. 1. Pandya-transformed matrix elements are indicated with a single bar

$$\bar{A}^{J}_{i\bar{l}k\bar{j}} = -\sum_{J'}(\hat{J}')^2\begin{Bmatrix} j_i & j_l & J \\ j_k & j_j & J'\end{Bmatrix} A^{J'}_{ijkl} \tag{8}$$

where the braces indicate the six-J coefficient, j_a denotes the angular momentum of the orbit a and we use the usual notation $\hat{J} \equiv \sqrt{2J+1}$. Cross-coupled matrix elements are indicated with a double bar

$$
\begin{aligned}
\overline{\overline{A}}^{J}_{j\bar{l}k\bar{i}} &= \sum_{J'} (\hat{J}')^2 \begin{Bmatrix} j_j & j_l & J \\ j_k & j_i & J' \end{Bmatrix} (-1)^{j_i+j_j-J'} A^{J'}_{ijkl} \\
\overline{\overline{A}}^{J}_{i\bar{k}l\bar{j}} &= \sum_{J'} (\hat{J}')^2 \begin{Bmatrix} j_i & j_k & J \\ j_l & j_j & J' \end{Bmatrix} (-1)^{j_k+j_l-J'} A^{J'}_{ijkl}
\end{aligned} \tag{9}
$$

In these definitions, we have not assumed Hermitian or permutation symmetries in the matrix elements A.

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Figure 1: Coupling schemes used in J-coupled expressions. (a) Standard coupling, (b) Pandya-transformed matrix elements, (c) cross-coupled matrix elements.

The one-body matrix elements are

$$f^{I}_{ij} = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abJ} \hat{J}^2 \cdot \chi^{\alpha}_{ab} \Gamma^{J}_{biaj} \tag{10a}$$

$$f^{II}_{ij} = \delta_{j_i j_j} \; \hat{j}_i^{-2} \sum_{abJ} \hat{J}^2 \left(\chi^{\beta}_{ab} - \chi^{\beta}_{ba} \right) \Omega^{J}_{biaj} \tag{10b}$$

$$f^{III_a}_{ij} = \delta_{j_i j_j} \; \hat{j}_i^{-2} \sum_{abcJ} \left(\bar{\chi}^{\gamma\ J}_{i\bar{c}a\bar{b}} \Gamma^{J}_{a\bar{b}j\bar{c}} - \bar{\chi}^{\gamma\ J}_{c\bar{j}a\bar{b}} \Gamma^{J}_{a\bar{b}c\bar{i}} \right) \tag{10c}$$

$$f^{III_b}_{ij} = \delta_{j_i j_j} \; \hat{j}_i^{-2} \sum_{abcJ} \left(\chi^{\delta\ J}_{ciab} \Gamma^{J}_{abcj} - \chi^{\delta\ J}_{abcj} \Gamma^{J}_{ciab} \right) \tag{10d}$$

with intermediates

$$\chi^{\alpha}_{ij} = \sum_{abcJ} \frac{\hat{J}^2}{2\hat{j}_i^2} \left(\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i \right) \Omega^{J}_{ciab} \Omega^{J}_{abcj} \tag{11a}$$

$$\chi^{\beta}_{ij} = \sum_{abcJ} \frac{\hat{J}^2}{2\hat{j}_i^2} \left(\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i \right) \Omega^{J}_{ciab} \Gamma^{J}_{abcj} \tag{11b}$$

$$\bar{\chi}^{\gamma\ J}_{i\bar{j}k\bar{l}} = \sum_{ab} \hat{J}^2 \left(n_a \bar{n}_b \bar{n}_k n_l - \bar{n}_a n_b n_k \bar{n}_l \right) \bar{\Omega}^{J}_{i\bar{j}a\bar{b}} \bar{\Omega}^{J}_{a\bar{b}k\bar{l}} \tag{11c}$$

$$\chi^{\delta\ J}_{ijkl} = \sum_{ab} \frac{\hat{J}^2}{4} \left(n_a n_b \bar{n}_k \bar{n}_l - \bar{n}_a \bar{n}_b n_k n_l \right) \Omega^{J}_{ijab} \Omega^{J}_{abkl} \tag{11d}$$

The J-coupled factorized expressions for the two-body matrix elements of the double commutator are

$$\Gamma^{I\ J}_{ijkl} = \sum_{a} \left\{ (1-\hat{P}^J_{ij}) \chi^{\epsilon}_{ai} \Gamma^{J}_{ajkl} + (1-\hat{P}^J_{kl}) \chi^{\epsilon}_{ak} \Gamma^{J}_{ijal} \right\} \tag{12a}$$

$$\Gamma^{II\ J}_{ijkl} = \sum_{a} \left\{ (1-\hat{P}^J_{ij}) \chi^{\zeta}_{aj} \Omega^{J}_{iakl} - (1-\hat{P}^J_{kl}) \chi^{\zeta}_{ak} \Omega^{J}_{ijal} \right\} \tag{12b}$$

$$\Gamma^{III_a J}_{ijkl} = \sum_{ab} \left\{ (1-\hat{P}^J_{ij}) \chi^{\eta\ J}_{ijab} \Gamma^{J}_{abkl} + (1-\hat{P}^J_{kl}) \Gamma^{J}_{ijab} \chi^{\eta\ J}_{abkl} \right\} \tag{12c}$$

$$
\begin{aligned}
\bar{\bar{\Gamma}}^{III_bJ}_{j\bar{l}k\bar{i}} =&(1-\hat{P}^J_{ij})(1-\hat{P}^J_{kl})\sum_{ab}\bar{\bar{\Gamma}}^J_{j\bar{l}a\bar{b}}\left(\bar{\bar{\chi}}^{\eta\ J}_{a\bar{b}k\bar{i}}+\bar{\bar{\chi}}^{\eta\ J}_{k\bar{i}b\bar{a}}\right) &(12d)\\
\bar{\Gamma}^{III_cJ}_{i\bar{l}k\bar{j}} =&\frac{1}{2}(1-\hat{P}^J_{ij})(1-\hat{P}^J_{kl})\sum_{ab}\bar{\chi}^{\theta\ J}_{i\bar{l}a\bar{b}}\bar{\Gamma}^J_{a\bar{b}k\bar{j}} &(12e)\\
\Gamma^{IV_aJ}_{ijkl} =&-\sum_{ab}\Big\{(1-\hat{P}^J_{ij})\chi^{\kappa\ J}_{ijab}\ \Omega^J_{abkl}\\
&+(1-\hat{P}^J_{kl})\Omega^J_{ijab}\ \chi^{\kappa\ J}_{klab}\Big\} &(12f)\\
\bar{\bar{\Gamma}}^{IV_bJ}_{j\bar{l}k\bar{i}} =&(1-\hat{P}^J_{ij})(1-\hat{P}^J_{kl})\sum_{ab}\bar{\bar{\Omega}}^J_{j\bar{l}a\bar{b}}\left(\ \bar{\chi}^{\iota\ J}_{a\bar{b}k\bar{i}}-\bar{\chi}^{\iota\ J}_{k\bar{i}a\bar{b}}\right) &(12g)\\
\bar{\Gamma}^{IV_c\ J}_{i\bar{l}k\bar{j}} =&\frac{1}{2}(1-\hat{P}^J_{ij})(1-\hat{P}^J_{kl})\sum_{ab}\bar{\chi}^{\lambda\ J}_{i\bar{l}a\bar{b}}\ \bar{\Omega}^J_{a\bar{k}j\bar{b}} &(12h)
\end{aligned}
$$

and the J-coupled intermediates are

$$
\begin{aligned}
\chi^{\epsilon}_{ij} =&\frac{1}{2\hat{j}_j^2}\sum_{abcJ}\hat{J}^2\left(\bar{n}_a\bar{n}_bn_c+n_an_b\bar{n}_c\right)\Omega^J_{ciab}\Omega^J_{abcj} &(13a)\\
\chi^{\zeta}_{ij} =&\frac{1}{2}\sum_{abcJ}\hat{J}^2j_j^{-2}\left(\bar{n}_a\bar{n}_bn_c+n_an_b\bar{n}_c\right)\\
&\times\Gamma^J_{aibc}\Omega^J_{bcaj} &(13b)\\
\bar{\chi}^{\eta\ J}_{i\bar{j}k\bar{l}} =&\sum_{ab}\left(\bar{n}_an_b\bar{n}_k+n_a\bar{n}_bn_k\right)\bar{\Omega}^J_{i\bar{j}a\bar{b}}\bar{\Omega}^J_{a\bar{b}k\bar{l}} &(13c)\\
\chi^{\theta\ J}_{ijkl} =&\sum_{ab}\left(n_an_b\bar{n}_k+\bar{n}_a\bar{n}_bn_k+n_an_b\bar{n}_j\right.\\
&\left.+\ \bar{n}_a\bar{n}_bn_j\right)\Omega^J_{ijab}\Omega^J_{abkl} &(13d)\\
\bar{\chi}^{\iota\ J}_{i\bar{j}k\bar{l}} =&\sum_{ab}\left(\bar{n}_an_b\bar{n}_k+n_a\bar{n}_bn_k\right)\bar{\Gamma}^J_{i\bar{j}a\bar{b}}\bar{\Omega}^J_{a\bar{b}k\bar{l}} &(13e)\\
\bar{\bar{\chi}}^{\kappa\ J}_{i\bar{j}k\bar{l}} =&\sum_{ab}\left(\bar{n}_an_bn_l+n_a\bar{n}_b\bar{n}_l\right)\bar{\bar{\Omega}}^J_{i\bar{j}b\bar{a}}\bar{\bar{\Gamma}}^J_{b\bar{a}k\bar{l}} &(13f)\\
\chi^{\lambda}_{ijkl} =&\sum_{ab}\left(\bar{n}_a\bar{n}_bn_l+n_an_b\bar{n}_l\right)\Gamma^J_{ijab}\Omega^J_{abkl}\\
&+\sum_{ab}\left(\bar{n}_a\bar{n}_bn_j+n_an_b\bar{n}_j\right)\Omega^J_{ijab}\Gamma^J_{abkl} &(13g)
\end{aligned}
$$

where $\hat{P}^J_{ij}\equiv(-1)^{j_i+j_j-J}P^J_{ij}$ is the J-coupled permutation operator.

We note that the intermediate operators appearing in equations (12) do not always have the same coupling scheme with which they are defined in equations (13). The translation between these can be obtained from equations (8) and (9).